PROJECT GUIDE & PRESENTATION COMPANION

AI-Based User and Entity Behaviour Analytics (UEBA) System

Learns how every user and device normally behaves, flags what is different, explains why, and runs as a secured system a security team can operate on real logs.

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Version 2 · October 2026 - covers the analytics, the operational system, the analyst workflow, security, and validation on the CMU CERT r4.2 benchmark.

CONTENTS

What is in this guide

Sections 1-3 tell the whole story in a few pages. Section 4 is the reference: every feature, what it does, **why it exists**, how it works and where it lives in the code. Sections 5-9 prepare you for the demo and the viva.

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> **The one-sentence pitch.** Stolen passwords and insiders look legitimate to signature tools; this system learns each person's and machine's normal behaviour, needs no attack labels, raises few alerts that explain themselves, and was tested honestly - on a simulated company it caught 12/12 attacks, and on the independent CERT benchmark it found 22 of 36 held-out insiders after one generic addition.

SECTION 1

The project in one page

A Python system that watches how every user and computer normally behaves and raises a short list of explained, risk-ranked alerts when something changes - and keeps doing so on its own.

The problem

Most breaches start with valid credentials or an insider (credential abuse is the top initial access vector in the 2025 Verizon DBIR; insider incidents cost US$17.4 M a year on average). Each log line looks legitimate, so signature tools miss them. Behaviour gives them away - a 3 a.m. login from abroad, ninety files instead of fifteen, gigabytes to a new site. The catch: behaviour-based tools drown analysts in false alarms and don't explain themselves.

What the system does

- **Captures** authentication, system-access, network / web-proxy and resource-usage logs - by upload, drop folder, authenticated API or an endpoint agent.
- **Profiles** 44 features per user-day (+7 for web activity) and 19 per device-day against each entity's own history and its department, adapting over time but never learning suspicious days.
- **Detects** deviations with three independent layers - robust statistics, an unsupervised ML ensemble (Isolation Forest, LOF, autoencoder) and 28 MITRE ATT&CK rules - fused into one 0-100 risk score that needs corroboration before it alerts.
- **Operates** continuously: a collector service scores only new data with saved models (identical to a full re-run), notifies analysts, groups alerts into cases, and the console is protected by roles and an audit trail.

Headline results

12 / 12	0.951	~1 / day	22 / 36
simulated attacks caught, each on day one	F1 (precision 0.935, recall 0.967; simulated)	alerts for 108 monitored entities	held-out CERT insiders (3 / 36 out of the box)

> **Be upfront about this.** The simulated company was designed together with the detector, so 0.951 is an upper bound. On CERT r4.2, which I did not design, the out-of-the-box system found only 3 of 36 held-out insiders; one generic indicator raised it to 22 of 36, and gradual job-seeker theft is still mostly missed. Saying this first is a strength in a viva, not a weakness.

SECTION 2

Key terms you must be able to explain

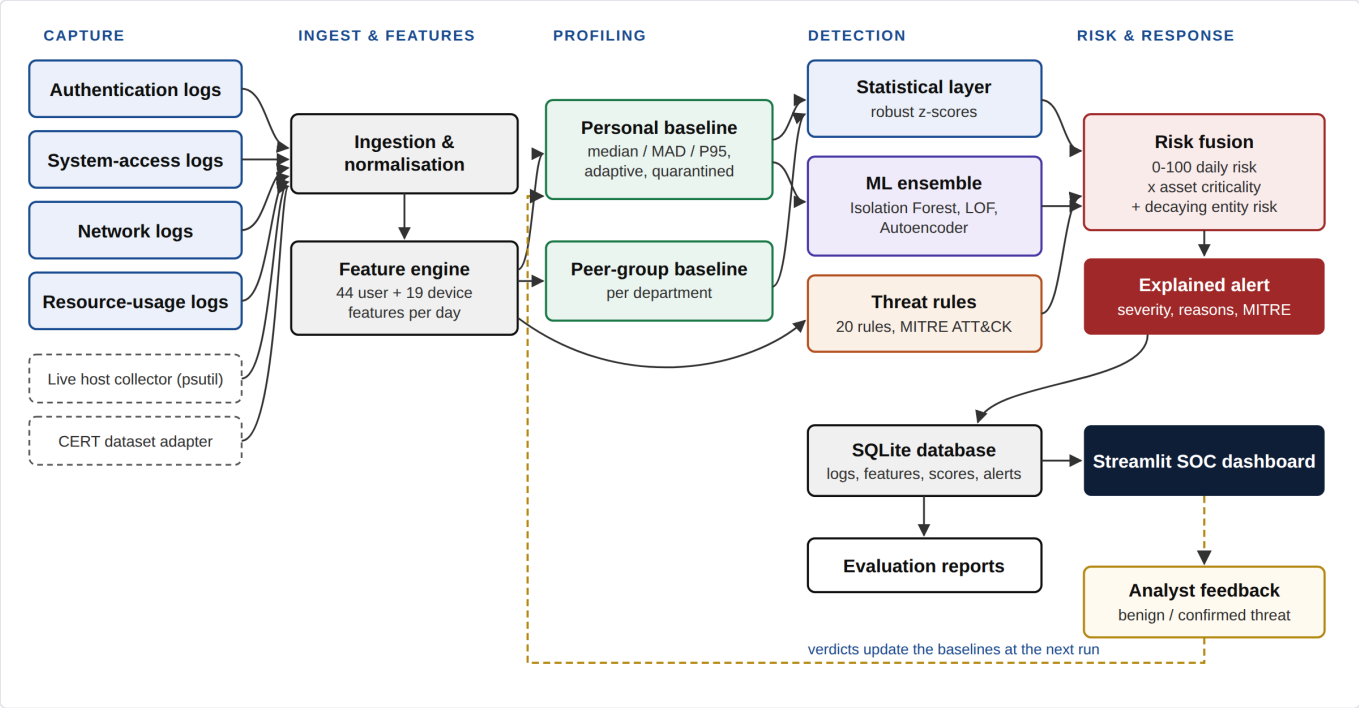
If you can explain this page in your own words, you can answer most "why" questions.

Term	Plain meaning, and where it shows up here
UEBA	User and Entity Behaviour Analytics: learn normal behaviour per user and per entity (devices, servers), flag deviations. Works without signatures, so it catches stolen credentials and insiders.
Entity-day	The unit of analysis: one user or device on one day, described by a feature vector.
Baseline / profile	What is normal for that entity: a centre (median) and a spread for every feature, plus usual hours, countries, hosts and domains.
Robust z-score	(today - median) / spread. "How many typical spreads away from normal". Uses the median and MAD (median absolute deviation) so one spike doesn't distort the baseline.
Peer group	The entity's department. A finance user who behaves like nobody in finance is suspicious even if they always did it.
Unsupervised	Learning without labels: the models only ever see normal days. Labels are used only to measure performance.
Isolation Forest	Random trees; unusual points are isolated in few splits.
LOF	Local Outlier Factor: a point in a sparse region compared with its neighbours is unusual.
Autoencoder	A neural network that compresses and rebuilds normal days (44-16-6-16-44); a day it rebuilds badly is unusual.
MITRE ATT&CK	The public catalogue of attacker techniques (T1110 brute force, T1567 exfiltration to cloud ...). Every rule carries its technique ID.
Precision / recall	Of the alerts raised, how many were real (precision - low means alert fatigue); of the real attacks, how many were alerted (recall).
ROC-AUC / PR-AUC	Threshold-free ranking quality. PR-AUC matters more when attacks are rare (~1% of days here).
Corroboration	Requiring two kinds of evidence (statistics + ML or a rule) before alerting.
Quarantine / poisoning	An attacker could slowly "teach" an adaptive baseline that the attack is normal (poisoning). Quarantined days are never learned.
Incremental scoring	Re-using trained models and scoring only new days instead of re-training on everything.
Held-out split	Data kept aside and never used for decisions, so results on it are honest.

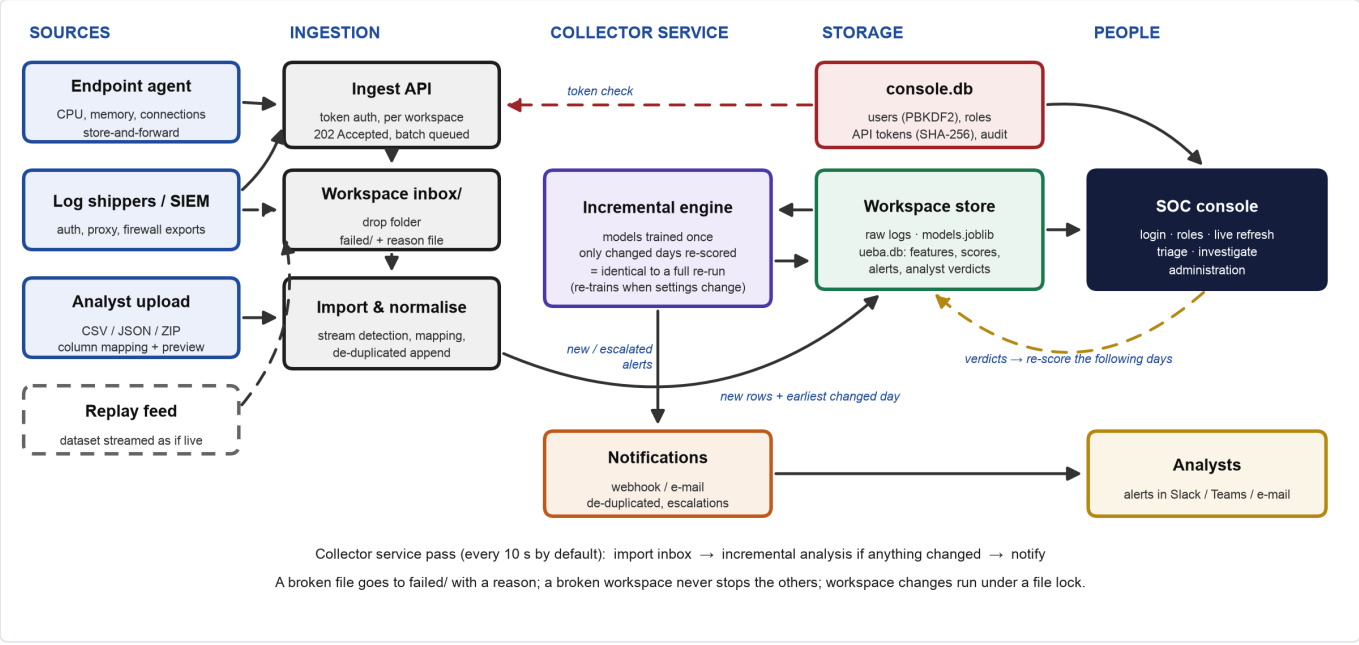
SECTION 3

Architecture

Two views of the same system: what happens to the data (analytics), and how it runs day to day (operation).



Analytics pipeline: capture → features → personal and peer profiles → three detection layers → risk fusion → explained alerts → storage, console and evaluation; analyst verdicts feed back into the baselines.



Operational system: logs arrive four ways and are queued in a workspace inbox; the collector service imports, scores incrementally and notifies; the console reads the results; console.db holds users, token hashes and the audit trail.

SECTION 4

Feature catalogue

Every feature has a reason to exist. Learn the **WHY** lines - they are what examiners ask about.

A · DATA IN

Four log streams

`ueba/ingest.py`

WHAT Authentication (logins, method, source IP, location), system access (files, commands, sensitivity), network and web proxy (domains, ports, bytes; with a proxy also the page, site category, allowed / blocked and downloaded file), resource usage (CPU, memory, disk).

WHY The problem statement asks for them, and attacks show up in different places: brute force in auth, theft in access and network, cryptomining in resource use.

HOW Validated, normalised to one schema and merged into one event timeline; missing streams are allowed.

Upload with automatic column mapping

`ueba/sources.py · Data sources page`

WHAT Drop CSV / JSON / ZIP exports; the log type is detected and columns mapped (TimeCreated → timestamp, TargetUserName → user); preview before import; the mapping is remembered.

WHY Real logs never use our column names. Without mapping the system could only analyse its own synthetic data.

HOW Synonym lists per field, checked against the column's values (URLs, IPs, dates); values normalised (Logon Failure / 4625 → failure, CORP\alice → alice, epoch times, CPU 0-1 → %); de-duplicated append. Linux auth.log / syslog text and Windows Security events (Event Viewer CSV, PowerShell, .evtx) are read directly; a file mixing event types is split by its type column.

Drop folder, ingest API and endpoint agent

`ueba/workspace.py · api.py · agent.py`

WHAT Files copied into a workspace inbox are imported automatically; agents and log shippers POST batches to a token-authenticated HTTP API; the agent streams a machine's CPU, memory and connections every 15 s.

WHY Nobody uploads logs by hand in production. Real data comes from export jobs, shippers and machines, continuously.

HOW The API only authenticates and queues into the inbox, so every route ends in the same import code. Bad files go to failed/ with a reason. The agent keeps data while the server is down (store-and-forward).

Workspaces and lifecycle

`ueba/workspace.py`

WHAT Independent datasets (Demo, CERT r4.2, your own) with their own logs, models, alerts and verdicts; status No data → New data → Learning → Monitoring.

WHY Mixing organisations would corrupt baselines; and a system that alerts before it has learned normal behaviour would only produce noise.

HOW A learning period (default 30 days) must be covered before alerts start; changes run under a file lock.

Replay as a live feed

`ueba/replay.py`

WHAT Streams a dataset into a workspace: 30 days of history, then one day every few seconds.

WHY To demonstrate and test the live system end to end - alerts appear as attacks happen.

Behavioural features (44 user, 19 device)

`ueba/features.py`

WHAT Daily counts such as failed logins, failures in 10 min followed by success, logins at hours rare for this user, new hosts / countries / domains, impossible travel, MB to new or cloud domains, USB use, CPU hours above 85%.

WHY Models can't read raw log lines; daily aggregates are cheap, private (metadata only) and explainable ("12 failed logins vs usual 1").

HOW "New" and "unusual hour" are computed only from earlier days (causal) - exactly what a live system sees.

Personal and peer baselines

`ueba/profiles.py`

WHAT Median and spread per feature for each entity and each department.

WHY "Unusual" only means something relative to a person's own normal; peers catch someone who was always odd for their role.

HOW spread = max(1.4826·MAD, (P95 - median)/1.645, floor). The P95 term respects personal peaks (a monthly 60 MB report) - it removed most false positives.

Adaptive baseline with quarantine

`ueba/pipeline.py`

WHAT Profiles are rebuilt daily from the last 45 accepted days.

WHY People change jobs and habits (stale baselines raise false alarms), but an attacker must not be able to teach the model that the attack is normal.

HOW Days with risk ≥ 60, any rule hit, or a confirmed-threat verdict are never learned; days an analyst clears as benign always are.

Three detection layers

`ueba/detectors.py`

WHAT Statistics (largest z-scores above 3), an unsupervised ML ensemble (Isolation Forest, LOF, autoencoder) trained on normal days only, and 28 rules mapped to ATT&CK.

WHY They fail differently: statistics explain, ML finds patterns nobody wrote a rule for, rules are precise for known techniques. No labels are needed - real organisations don't have them.

HOW Each ML score is calibrated so 1.0 = the 99th percentile of normal, then averaged.

Web activity: what the user did online

`ueba/web.py · features.py · detectors.py`

WHAT From proxy logs (Squid, web-gateway CSV): pages visited, blocked requests and blocked sites, visits to hacking / anonymizer / malware sites, kinds of site never visited before, downloads and executable downloads - 7 more features per user-day - plus 4 rules: a burst of blocks vs the person's usual rate, a proxy / VPN site within an hour of a block (filter bypass, T1090), an executable from a never-visited site (T1105), first visits to hacking sites (T1588.002).

WHY The review asked what a person actually did, not only how much data moved. Being blocked, getting round the filter and fetching tools are early signs that byte counts miss.

HOW The site's category comes from the gateway or a built-in list; "blocked" from TCP_DENIED / 403 / deny; the file from the URL. The features join the models only when the logs carry this detail, so flow-only data (CERT) is scored exactly as before. Profiling it raised the ML ensemble alone from F1 0.79 to 0.86.

MORE Subdomains: each request keeps its host (erp.ibm.com) under the organisation's domain (ibm.com); a never-used subdomain is flagged. Time spent: requests to a site under 5 minutes apart form one visit (first to last request + 1 min), giving minutes per website and minutes online vs usual. Two rules use them: hundreds of random-looking subdomains of one site (tunnelling, T1071.004) and a new site hit at clockwork intervals all day (malware beaconing, T1071.001).

Rule R16 - removable media without history

ueba/detectors.py

WHAT USB use by an account that never used removable media in its learning period (T1052.001).

WHY Added after CERT showed quiet insiders slip through; a standard insider indicator. Chosen on a development half only; raised held-out detection from 3/36 to 22/36; never fires on the simulated company.

Risk fusion and corroboration

ueba/risk.py

WHAT A = 1 - (1-0.28·stat)(1-0.10·peer)(1-0.30·ml)(1-rules); daily risk = 100·min(1, A·criticality).

WHY One number to rank by, weighted by impact (admins, servers ×1.3). The threshold 45 is above what statistics alone can reach (43.8), so **an alert always needs ML or a rule to agree** - this raised precision from 0.31 to above 0.9.

HOW Behaves like a probabilistic OR: one strong signal or several weak ones raise risk; never above 100.

Accumulated risk

ueba/risk.py

WHAT entity risk(t) = entity risk(t-1)·0.5^(days/3) + max(0, daily risk - 15); a "sustained risk" alert at 60.

WHY Catches attackers who stay a little suspicious for several days without one alarming day.

Plain-English explanations

ueba/detectors.py

WHAT "3,300 MB uploaded vs usual 1.2 (2750× usual) - also unusual vs peers", layer scores and ATT&CK IDs on every alert.

WHY Analysts mistrust scores they can't reason about; an explained alert is closed or escalated in seconds.

C · RUNNING CONTINUOUSLY

Incremental engine

ueba/pipeline.py

WHAT Models are trained once and saved; later analyses score only the days from the earliest changed day.

WHY Re-training on all history at every update doesn't scale, and would silently change past scores.

HOW Because every feature is causal, the result is **identical to a full re-run** - an automated test cuts the data at a day boundary and mid-day and checks. Saved models carry a fingerprint of the settings; changing settings, the user directory or the learning period triggers re-training automatically.

Collector service

ueba/service.py · launch.py

WHAT Every 10 s: import the inbox → incremental analysis if anything changed → notify. Writes a heartbeat the console shows.

WHY Detection only helps if it happens without someone clicking "analyse".

HOW One broken file or workspace never stops the others; ~2-3 s per update.

Notifications

ueba/notify.py

WHAT Webhook (Slack, Teams, Discord, any JSON endpoint) or e-mail for new or escalated alerts.

WHY Nobody watches a dashboard all day.

HOW Each alert sent once (again only if it escalates); closed alerts and imported history stay quiet; every attempt logged.

D · ANALYST WORKFLOW

Cases

`ueba/cases.py · Alerts page`

WHAT Alerts on one entity within 7 days form one case with status, assignee and a notes timeline.

WHY An analyst should handle a person once, not 30 times; on CERT, 424 alerts became 195 cases.

HOW Closing a case records a verdict for every day it covers, so the baseline learns or quarantines them.

Verdicts that teach the system

`ueba/feedback.py`

WHAT Benign / threat on an alert or a case.

WHY The same legitimate behaviour shouldn't be flagged again; a confirmed attack must never become "normal".

HOW The days after a verdict are re-scored automatically with the corrected baseline.

Suppression

`ueba/cases.py`

WHAT Silence one alert type for one entity (or everyone) until an expiry date, with a required reason.

WHY Known patterns (a night-shift backup admin) cause alert fatigue; on CERT one habit produced 91 alerts for one user.

HOW Suppressed alerts are still computed and kept, hidden and never notified; audited; can be lifted.

User activity page

`dashboard/views/activity.py`

WHAT For any person and day: a Websites tab (each site, its subdomains, minutes, visits, data in / out, flags, minutes vs usual) and one story of logins, pages, downloads, blocked attempts and files, routine repeats folded, and every step that is new or risky for that person flagged; tabs per kind; today vs usual numbers and kinds of site.

WHY A risk score says how unusual a day was; an analyst (and a panel) needs to see what happened. Opened from any alert with "What they did".

Investigation, performance and log pages

`dashboard/views/`

WHAT Entity page (learned baseline, risk timeline, biggest deviations, event timeline); model performance with a what-if threshold; per-workspace alert threshold; log explorer with CSV export; live refresh.

WHY Context in one place makes triage fast; each organisation must tune its own threshold (CERT proved it).

E · SECURING THE TOOL ITSELF

Accounts, roles and passwords

`ueba/auth.py · login view`

WHAT Viewer (read), analyst (+ verdicts, cases), administrator (+ data, settings, users, tokens); users can change their own password.

WHY The console holds an organisation's security logs - it is itself a target.

HOW PBKDF2-HMAC-SHA256 (310,000 iterations, salt per user), constant-time compare, generic error messages, lock-out after 5 failures for 15 min, 8-hour sessions.

API tokens and the audit trail

`ueba/auth.py · Administration page`

WHAT 256-bit tokens shown once, stored only as SHA-256, scoped to one workspace, revocable; every login, failed login, verdict, upload, setting, user and token change is recorded.

WHY Agents must authenticate without sharing a user's password; actions on security data must be accountable.

Safe defaults `.streamlit/config.toml · ueba/service.py`

WHAT Console and API listen on this machine only; a team deployment goes behind a TLS reverse proxy.

WHY Streamlit would otherwise be reachable by anyone on the same Wi-Fi.

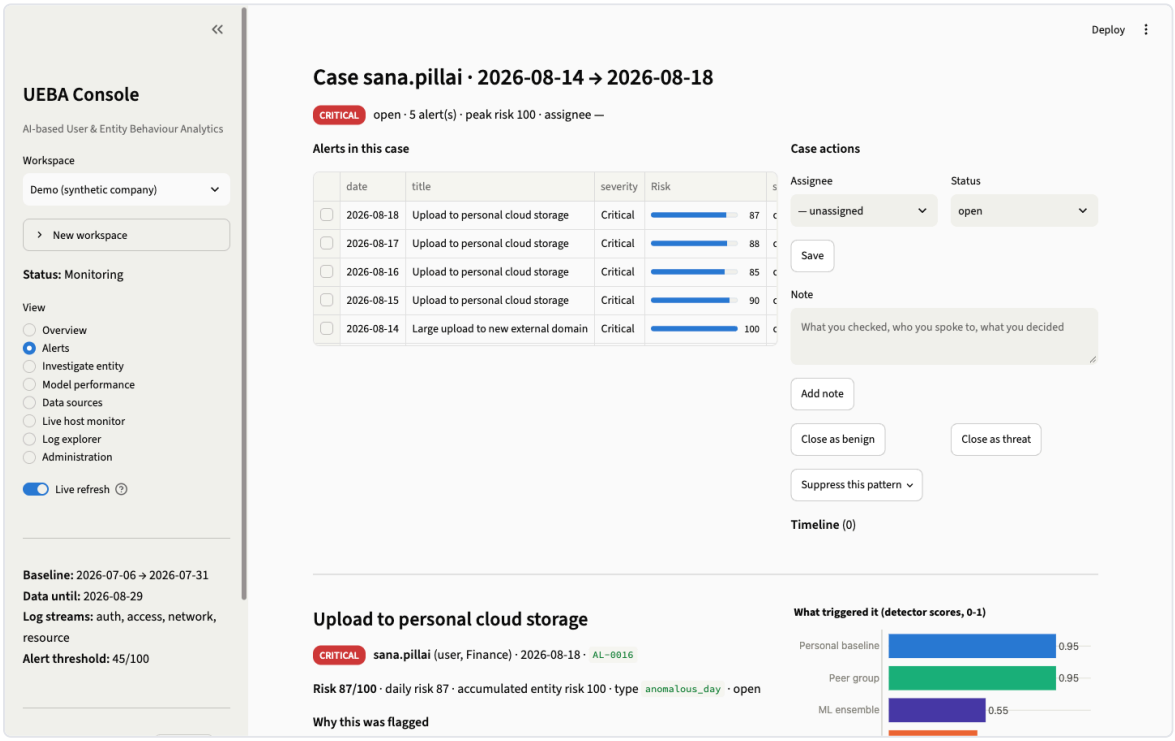
SECTION 5

Dashboard tour

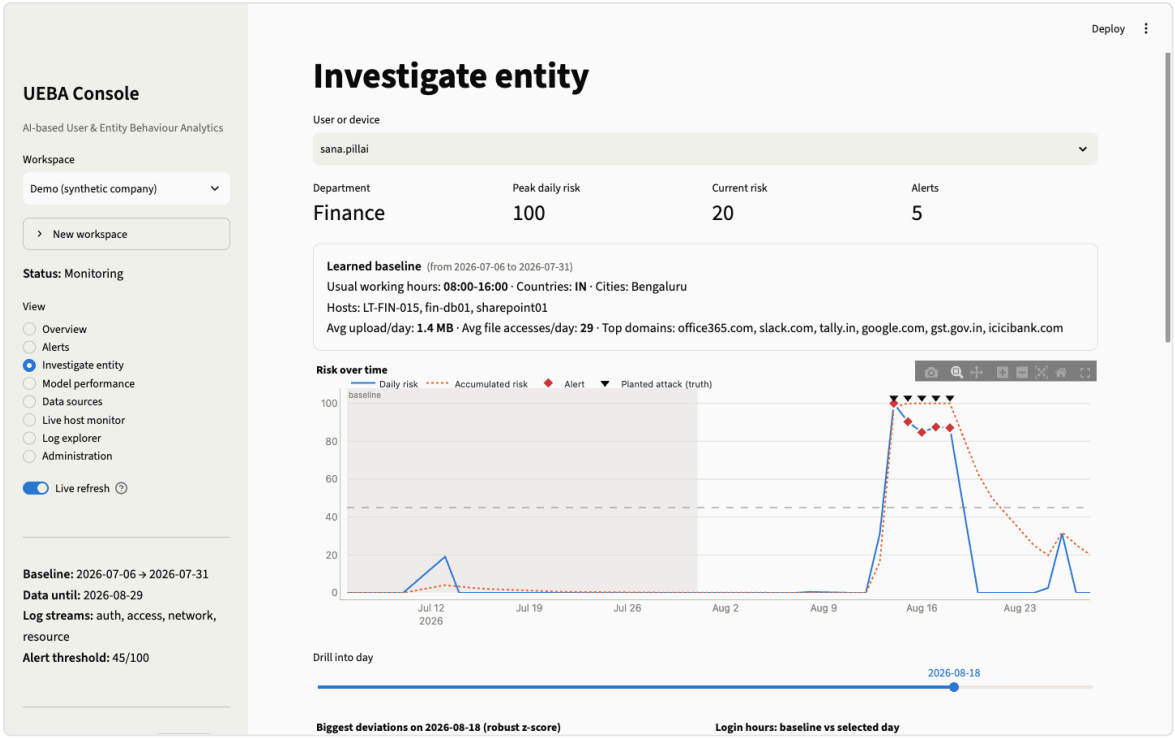
Nine pages, ordered the way an analyst works. Each workspace shows its status and the collector service's health in the sidebar.



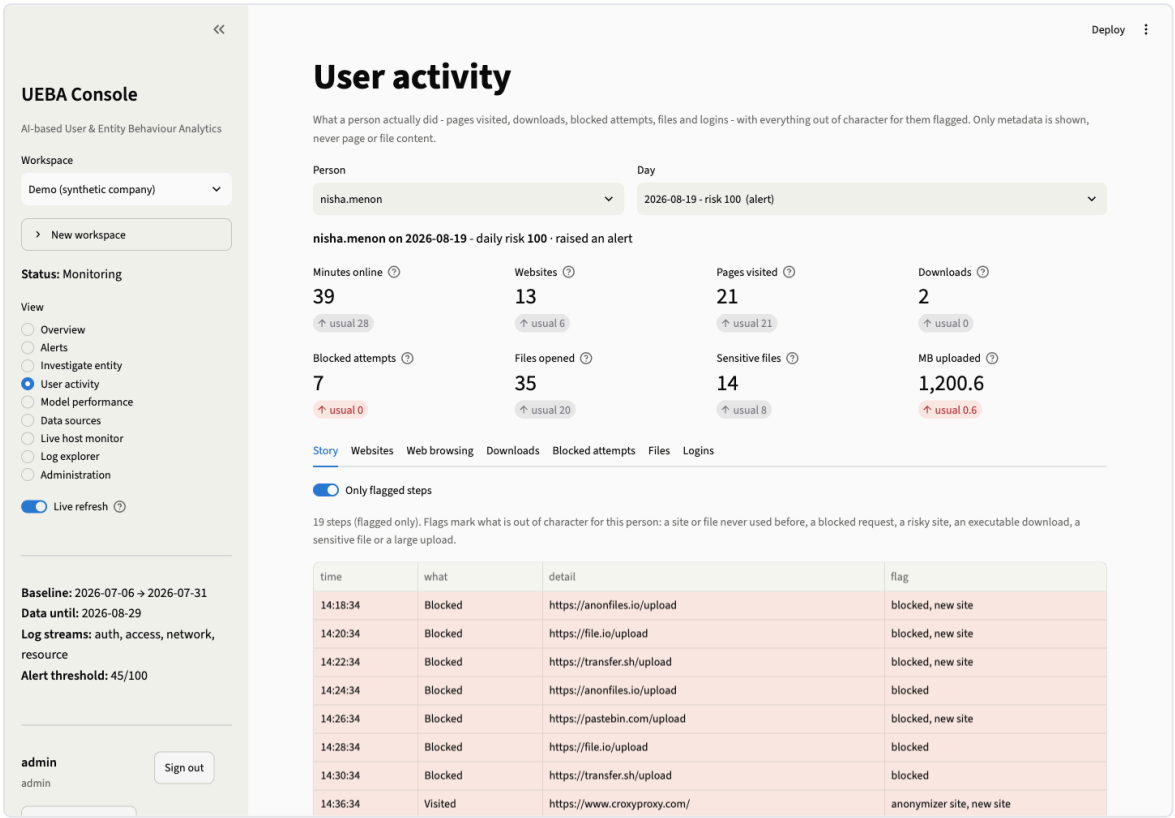
Overview - events, entities, alerts, open cases, alerts per day (weekly for long periods), riskiest entities, ATT&CK coverage, risk by department.



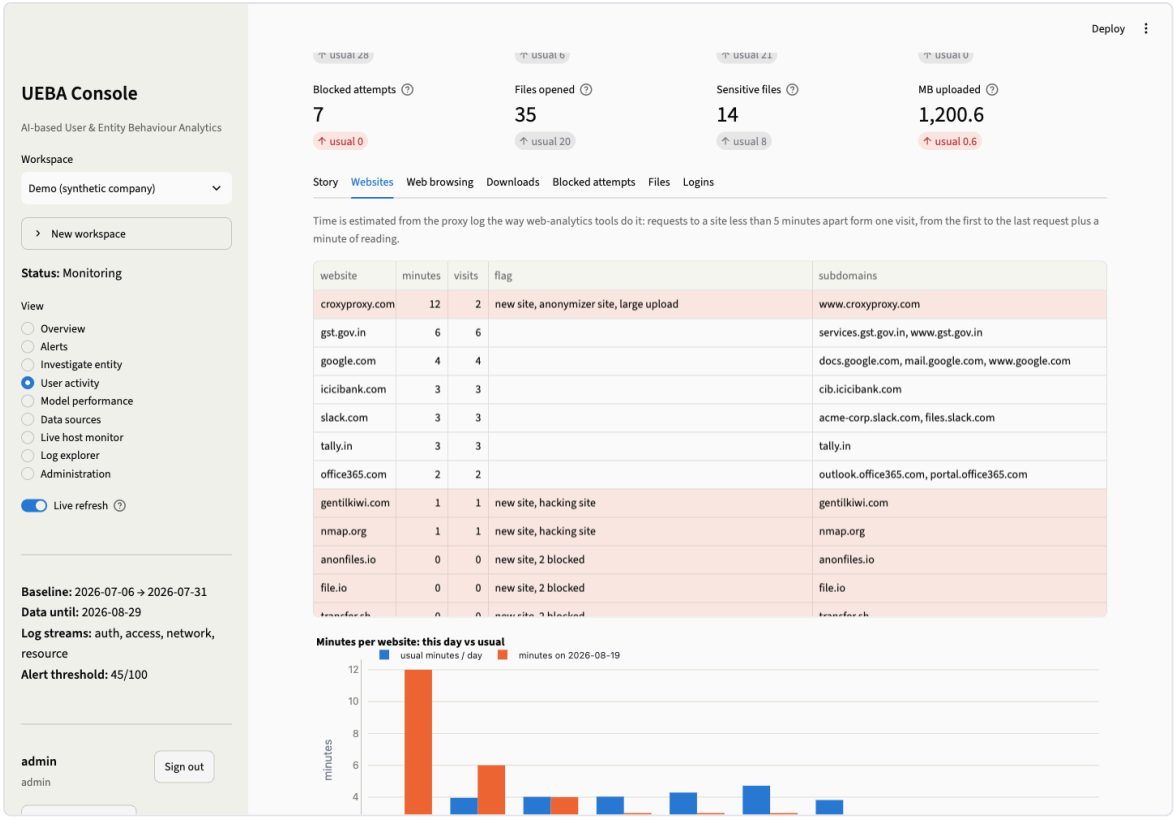
Alerts → Cases - the slow insider's five daily alerts as one case: assign, set status, add notes, close as benign or threat, suppress a pattern; the explained alert below.



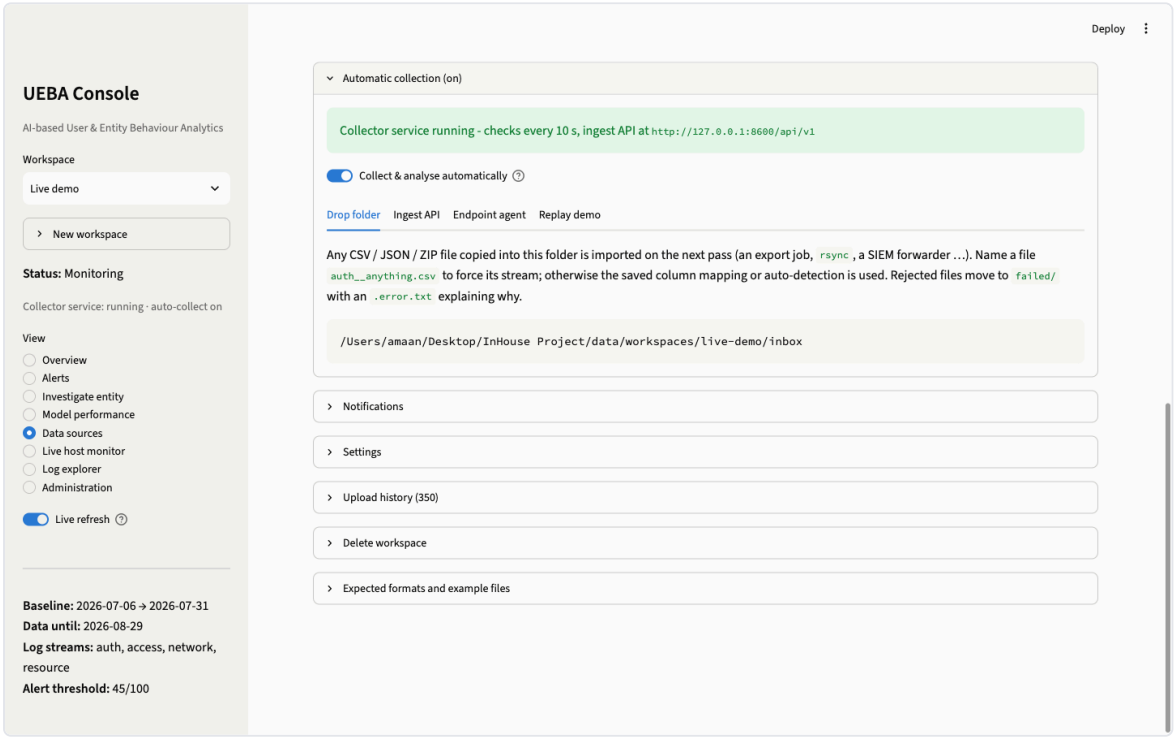
Investigate entity - learned baseline (hours, countries, hosts, volumes), risk over time with alerts and the labelled attack days, then a drill-down into any day.



User activity - the web-filter evasion as a flagged story: seven blocks, a proxy site, nmap and mimikatz downloads, first-time finance files, then 1.2 GB uploaded through the proxy.



User activity → Websites - every site with its subdomains (docs. / mail. / www.google.com), estimated minutes and visits, data in / out and flags; minutes per site compared with the person's usual.



Data sources → Automatic collection - service status, the drop folder, the ingest API, the agent and replay commands; notifications and settings below.

Page	Use it to
Model performance	See precision / recall / ROC with a what-if threshold; apply a new threshold
Live host monitor	Take a one-off sample of this machine (the agent does it continuously)
Log explorer	Browse any raw stream by user, device and date; export CSV

Administration	Users and roles, API tokens, audit log, service health (administrators only)

SECTION 6

Results

Simulated organisation (48 people, 12 servers, 60 days, 12 attacks)

Measure	Value
Incidents detected	12 / 12, each on the first day
Precision / recall / F1	0.935 / 0.967 / 0.951 (TP 29, FP 2, FN 1)
False positive rate · alerts per day	0.04 % · about 0.9
Five different organisations (mean F1)	0.945, 60 / 60 incidents
Behavioural layers only, no rules (F1)	0.854

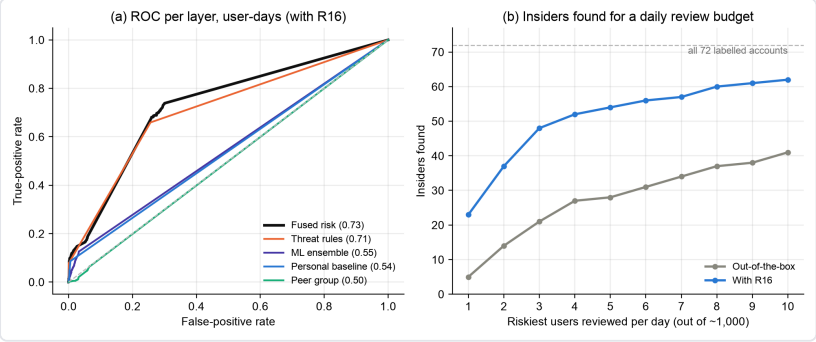
The one false positive is a planted harmless oddity (a new project, risk 49); the one false negative is the pivot laptop of the lateral movement (43.6, just below 45). Rules alone score F1 0.983 only because they were written for these attack types - the behavioural layers reach 0.854 without knowing the attacks.

Live operation

Incremental updates gave **identical** scores and alerts to a full re-run (cut at a day boundary and mid-day). Replaying the company as a live feed (8-hour slices): 1 training + 89 incremental updates of 1.8-5.7 s (median 2.6), no errors, 12/12 incidents and F1 0.951 - the same as the batch run.

CERT r4.2 (1,000 users, 70 insiders, 17 months) - held-out half

Measure	Out of the box	With R16
Insiders alerted on a malicious day	3 / 36	22 / 36
· scenario 1 (new USB user leaks data)	1 / 18	18 / 18
· scenario 2 (job seeker steals data)	1 / 14	1 / 14
· scenario 3 (admin, keylogger)	1 / 4	3 / 4
Alerts per day (~500 users)	0.66	0.84
Insiders among the 3 riskiest users per day	12 / 36	28 / 36



(a) The statistical and ML layers rank CERT days barely above chance - they ignore small deviations. (b) Insiders found for a daily review budget.

What did not work - and why that's fine to say. A 7-day trend layer for gradual insiders raised ROC-AUC by ~0.1 but either caught nobody new at ~1 alert a day, or caught half the job seekers at 11× the alerts, and it made the analysts' top-5 list worse. It was rejected using the development half only. Gradual insiders need sequence models (LSTM, graph-sequence work in the literature review).

SECTION 7

How to run it, and the code map

```
python3 -m venv .venv && source .venv/bin/activate && pip install -r requirements.txt
python run_pipeline.py                # build the demo company and analyse it (~30 s)
python launch.py                      # console on http://localhost:8501 + collector service +
ingest API
python -m ueba.replay --workspace live-demo --labels      # stream the company live into a workspace
python -m ueba.agent --url http://127.0.0.1:8600 --workspace lab --token ueba_...
python run_pipeline.py --cert data/cert/r4.2              # CERT benchmark into its own workspace (~8
min)
python experiments.py · experiments_cert.py · experiments_trend.py   # robustness / CERT / trend
experiments
python -m pytest -q                   # 35 tests
```

File	Responsibility
`ueba/generator.py`	Simulated company, planted attacks, ground truth
`ueba/sources.py · ingest.py`	Log-type detection, column mapping, normalisation, storage
`ueba/features.py · profiles.py`	Behavioural features; personal and peer baselines
`ueba/detectors.py · risk.py`	Statistics, ML ensemble, 28 rules, explanations; risk fusion, alerts
`ueba/pipeline.py`	Full training run or incremental update; adaptive baseline with quarantine
`ueba/workspace.py`	Workspaces, inbox, locking, learning → monitoring lifecycle
`ueba/service.py · api.py · agent.py · replay.py`	Collector service; ingest API; endpoint agent; live replay
`ueba/cases.py · feedback.py · notify.py`	Cases and suppressions; verdicts; webhook / e-mail
`ueba/auth.py`	Users, roles, password hashing, API tokens, audit log
`ueba/cert_loader.py · evaluate.py`	CERT adapter; metrics, figures, reports
`dashboard/app.py · ui.py · views/`	Console shell; per-session context; one module per page
`tests/`	35 tests: detection, ingestion, web activity, incremental equivalence, API, security, cases, service

SECTION 8

Five-minute demo script

Before the viva: run `python launch.py`, sign in, create a workspace "Viva live", add a webhook (e.g. a webhook.site URL open on your phone), and keep a terminal ready with `python -m ueba.replay --workspace viva-live --seconds-per-day 8 --labels`.

Time	Show	Say
0:00	Sign-in screen	"A security tool must be secure itself - roles, hashed passwords, lock-out, and everything is audited."
0:30	Demo → Overview	"A simulated 48-person company, 206k events, 12 attacks and 4 harmless oddities. All 12 caught on day one, two false alerts in 30 days."
1:15	Alerts → Cases → sana.pillai	"Five daily alerts of a slow insider are one case. Every alert explains itself, and statistics alone can never alert - ML or a rule must agree."
2:00	Investigate →	"Her learned baseline, the risk timeline, and exactly what deviated."
2:15	Alerts → nisha.menon → What they did	"And this is what a person actually did: blocked seven times on file-sharing sites, switched to a web proxy, downloaded nmap and mimikatz, then sent 1.2 GB out."
2:30	Start the replay; open Viva live	"Now as a live system - history arrives, it learns, then a day every 8 seconds. Nobody clicks anything."
3:30	INC-01 appears; phone buzzes	"Brute force from Tor - detected and notified within one service pass. Models were trained once; each update scores only new data, provably identical to a full re-run."
4:15	CERT figure	"On CMU's benchmark, which I didn't design: 3 of 36 out of the box; one generic indicator chosen on a separate half - 22 of 36. Job-seekers are still mostly missed; that's my future work."
4:45	Data sources → Automatic collection	"Real logs come in by upload with automatic mapping, a drop folder, an authenticated API, or the agent."

SECTION 9

Viva questions and answers

Short answers you can say in 20-30 seconds. Understand them; don't recite them.

Why unsupervised, not a classifier?
Real organisations have almost no labelled attacks, and the next attack is not like the last one. The models learn only normal behaviour; labels are used only to measure.

Your results come from data you generated - isn't that circular?
Yes, it's the main limitation. That's why I ran CERT, which CMU built: 3/36 out of the box, 22/36 with one generic indicator chosen on a separate half. The synthetic 0.951 is an upper bound.

Isn't adding R16 after seeing CERT tuning on the test set?
I split the users first, chose R16 on the development half (4 → 17) and quote the untouched test half (3 → 22). It's a standard indicator and never fires on the synthetic data. The out-of-the-box number is reported as the generalisation result.

Rules alone score 0.98 - why the ML?
Because I wrote those rules for the attacks I also simulated. Without rules the behavioural layers still reach F1 0.854 with no knowledge of the attacks.

Can an attacker poison the baseline?
Risky days (≥ 60), rule hits and confirmed threats are never learned; the ML models are trained once on the learning period.

Why median and MAD, not mean and standard deviation?
One legitimate spike would inflate a standard deviation and hide the next anomaly; the median ignores outliers, and the P95 term respects personal peaks.

How is incremental scoring correct?
All features are causal, so a day's score never depends on later data; the engine reuses saved models and earlier accept / quarantine decisions and re-scores from the first changed day. A test checks equality with a full re-run.

Why so much worse than published CERT results?
Those models use labels or per-session sequences; mine uses no labels, daily aggregates, and thresholds calibrated elsewhere. The fix is per-organisation calibration and sequence-aware scoring.

How do you avoid alert fatigue?
Corroboration threshold, explanations, cases instead of repeated alerts, suppression with expiry, notifications only for new or escalated alerts.

What would a real deployment need?
TLS via a reverse proxy, the service as a system service, native EVTX / syslog parsers, PostgreSQL at scale, and SOAR for automated response.

Limitations to state before they are asked

- Quiet, gradual insiders (CERT scenario 2) are mostly missed; the behavioural layers ignore small deviations.
- Synthetic results are an upper bound; real logs are messier.
- Daily granularity: alerts within seconds of data arriving, but no intra-day sequence detection.
- The agent can't measure bytes per connection; upload volumes and pages come from proxy / firewall logs, and HTTPS pages are visible only where the proxy inspects TLS.

> **Cheat sheet.** 205,881 events · 108 entities · 44 (+7 web) + 19 features · 3 layers · 28 rules · threshold 45 (stats max 43.8) · F1 0.951 (P 0.935) · F1 0.945 over 5 seeds · 0.854 without rules · CERT 3 → 22 of 36 · ~1 alert/day · update ~2.6 s · 35 tests.